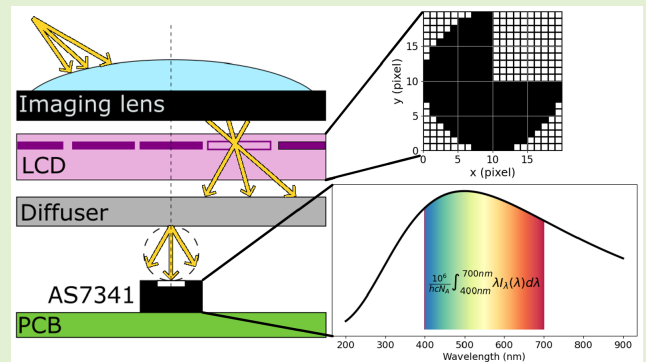


A Low-Power Optical Sensor With Dynamically Adjustable Field of View for Photosynthetically Active Radiation (PAR) Measurement

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Abstract—The characterization of light as photosynthetically active radiation (PAR) is crucial for modeling and monitoring plant metabolism and development. Existing intercepted PAR sensing solutions are limited by the requirement to average many below-canopy measurements to account for heterogeneity in the plant canopy, which requires a human operator. In this work, we address this issue by introducing a novel sensing concept to measure the light coming from different areas of the tree canopy with a single sensor by dynamically varying the sensor's field of view (FOV) in the zenith and azimuth directions. The sensing system consists of: 1) an imaging lens that collects the incoming light; 2) a programmable liquid-crystal display (LCD), which acts as an optical shutter; 3) a diffuser; and 4) the AS7341 multispectral sensor. In this work, the PAR model required to calibrate the sensor is derived and the sensing system is calibrated and characterized. A sensor prototype is then assembled, calibrated, and characterized, achieving competitive performance with a mean average percentage error of 6.78% and a normalized root mean square error of 4.82% on PAR measurements. Tests using a custom LED test sphere show that the sensor has a spatial resolution of 7° and 15° in the θ - and ϕ -directions, respectively. With a power consumption of 19.54 mW mainly attributable to the LCD, our new sensing system could be used for more than two years at one hour of measurements per day, which paves the way toward long-term continuous monitoring of PAR.

Index Terms—Liquid-crystal display (LCD), long term deployment, photodiode array, photosynthetically active radiation (PAR), sensor node.



I. INTRODUCTION

UNDERSTANDING and characterizing plant metabolism is crucial to predict and mitigate the impacts of the ongoing climate change [1]. Incident light is one of many

important parameters affecting plant metabolism because of its role in photosynthesis: light energy is absorbed by chloroplasts and converted into chemical energy in the form of sugar or other organic molecules [2].

The light used for photosynthesis is referred to as photosynthetically active radiation (PAR) and is quantified as the photon flux per ground area per second ($\mu\text{mol m}^{-2} \text{s}^{-1}$) in the PAR (400–700 nm) [3]. Of particular interest to researchers is the PAR intercepted by the canopy, which is obtained by subtracting a below-canopy PAR measurement from an above-canopy measurement. This can be done using one sensor with the two measurements realized in quick succession or two sensors measuring the PAR simultaneously above and below the canopy. This intercepted PAR strongly correlates with plant growth and production of dry matter [4].

PAR can be measured with commercial sensors based on one or multiple photodiodes. One example is the LI-190R Quantum Sensor (LI-COR, Nebraska, USA), consisting of a silicon (Si) photodiode covered by a custom spectral filter and a cosine-corrected acrylic diffuser [5]. The photodiode current

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is directly proportional to the PAR. Up to three sensors can be connected simultaneously to the LI-1500 Light Sensor Logger (LI-COR, Nebraska, USA), which converts the photodiode current to the PAR using a calibration constant.

Recent developments in remote sensing have enabled the estimation of PAR from satellite data. PAR values can be retrieved with a 1–5 km resolution and time scales of hours to days [6], [7]. While this is a cost-effective solution to obtain PAR values over a wide area, it is not appropriate for intercepted PAR measurements for small groups of trees or single trees because of the limited spatial resolution and inaccuracies due to dynamic cloud coverage.

Since intercepted PAR is affected by heterogeneity in the canopy, it is recommended to use the average of measurements taken at different positions under the canopy [8]. In the case of the LI-190R Quantum Sensor, an operator must take many measurements at different positions, which prevents use in a continuous wireless monitoring setting. Another commercial sensor, the AccuPAR LP-80 (METER Group, USA), partially addresses this issue using a 87 cm-long measurement probe for below-canopy PAR measurements. The probe is composed of eight sensor segments that can be activated or not to account for canopy heterogeneity and leaf clumping [9]. While appropriate for monitoring a single tree, this commercial solution is not appropriate to the simultaneous monitoring of multiple trees, because many exemplars of this expensive sensing system would need to be used in parallel.

As continuous unassisted monitoring of the PAR enables precise modeling of plant metabolism and establishing long-term correlations, many PAR sensing solutions have been developed to enable long-term monitoring. Some sensing solutions integrate a commercial sensor in a custom sensor node [10], [11], [12], and others use a custom quantum sensor [13]. However, none of those solutions address the handling of heterogeneous canopies: they are not able to take measurements in different positions under the canopy automatically with one single sensor.

To overcome these limitations, we present in this work an innovative light sensor for which the incident field of view (FOV) could be dynamically varied in the azimuth and zenith directions. This would enable the sensor to scan the canopy and thus identify and properly handle the heterogeneity in the canopy. Many technologies have been considered to perform this FOV restriction, and the most suitable in terms of cost, power consumption, and robustness to prolonged usage in nature have been identified. Specific portions of the incoming light can be blocked with a mechanical shutter or an optical shutter. Alternatively, instead of blocking light coming from specific directions, the system could use a scanning mechanism combined with a narrow FOV to scan a desired area of the sky.

Mechanical shutters are most present in professional camera systems to control the exposure of the photographic film or sensor [14]. However, the FOV delimited by their shutters cannot be precisely set: the shutter mechanism is first triggered and then reset. Therefore, these shutter systems are not appropriate for a context where the position of the shutter should be precisely set and held for a certain amount of time. Aside from these macro-mechanical shutters, micro-mechanical shutters

have been developed in recent years for applications such as smart glass technology. In their review of microshutter technology, Lamontagne et al. [15] find that microshutters have adequate performance and can be implemented on a small scale, but that more work is required for reliable large-scale use and manufacturing. The required actuation voltage can be up to 100 V [16].

Optical shutters are shutters that do not depend on mechanical components. They usually rely on manipulating a fluid to selectively transmit, absorb, or reflect light incident onto the system. The most well-known technology is the use of liquid crystals for display applications, commonly called liquid crystal displays (LCDs). Applying a voltage across a liquid crystal cell rearranges the long molecules inside the fluid such that the light polarization state is modified. By placing crossed linear polarizers before and after the cell, the applied voltage controls whether light is transmitted or not [17]. LCDs must be actuated with a continuous ac voltage signal of ≈ 100 Hz with amplitude of 5–10 V. The power required to actuate a liquid crystal cell is $\approx 500 \mu\text{Wcm}^{-2}$ [18]. Liquid crystals are already used in shutter technology, such as in Thorlabs' Liquid Crystal Optical Beam Shutter/Variable Attenuator (Thorlabs, New Jersey, USA). Whereas LCDs require constant power to maintain a given state, smart windows' technology such as electrochromic (EC) windows require no power at all. Widely used EC materials are oxide-based, and their transmittance is varied through controlled reduction–oxidation reactions [19]. The application of a voltage controls the redox reaction via the transport of ions from the ion storage film to the EC film. For a given foil-based device, the transmittance can be modulated from 70 % down to 15 % in 30 s [19]. The required voltage is typically not more than 10 Vdc [18], [19]. EC-based displays have been prototyped for more than 30 years, but they have never established themselves because of slow response times and lack of a sharp switching threshold [20]. The technology is still developed to this day due to its very low power and ability to be used on flexible substrates, which is an advantage compared to mainstream display technologies [21]. Aside from display technologies, microfluidic technology can be used to control the position and spread of a light-absorbing liquid. One example consists of the use of two liquids (one transparent and one opaque) whose ring-shaped liquid–liquid interface can be deformed using electrowetting [22]. This enables dynamically adjusting the diameter of the iris. This requires a 75 V drive voltage and has a power consumption of 5 mW.

Another way to dynamically vary the FOV of a sensor is to use a sensor with limited FOV in combination with a scanning mechanism, such that the sensor can scan the desired FOV “point-by-point.” This principle is used in a leaf area index (LAI) sensing solution by Wen et al. [23], where servomotors orient the sensor toward a desired position. Another ubiquitous scanning technology is the digital micromirror device (DMD) from Texas Instruments (Dallas, USA). The DMD is composed of an array of micromirrors whose orientation can be controlled, usually with two positions [24]. In an application of light sensing, incoming light could be either reflected into the sensor or toward a light absorber on a pixel-by-pixel basis. The smallest DMD produced by Texas Instruments is the

DLP2021-Q1 with $588 \text{ pixel} \times 330 \text{ pixel}$ and maximum power consumption of 260 mW [25].

The LCD is selected for this work because of its low power consumption, wide commercial availability, and robustness. The proposed solution builds upon an existing low-power PAR and LAI sensor node developed by Comella et al. [26], [27], which uses a multispectral sensor as the light sensing apparatus. The dynamic FOV variation is achieved through the combination of imaging optics and a transparent liquid-crystal display (LCD) placed in front of the multispectral sensor [28]. The combination of a multispectral light sensor and the ability to control the incoming direction of the measured light has the potential to mitigate the impact of leaf clumping and non-leaf components on the PAR measurements. Furthermore, this development would allow the simultaneous measurement of both PAR and light spectrum composition with a single sensor.

The main contributions of this work are as follows:

- 1) the introduction of the novel sensor concept and its key components;
- 2) the derivation of a PAR model which takes into consideration the impact of each component in the path of the incident light;
- 3) the conceptualization and implementation of an appropriate calibration procedure for the sensor, including the use of a novel LED test sphere setup;
- 4) the characterization of the sensor's spatial resolution and power consumption.

This article is organized as follows. Section II describes the novel sensor concept and its components. Section III presents the derivation of the PAR measurement model. This model then guides the selection of components for the realization of a sensor prototype. This prototype along with the calibration and characterization methods are presented in Section IV. The calibration and characterization results are shown and discussed in Section V. Section VI concludes this publication by summarizing key takeaways and highlighting avenues for future work.

II. SENSOR CONCEPT

The sensor concept builds on a previously developed PAR and LAI sensor node [26], [27] by adding the possibility to restrict the FOV to a specific area of the sky using an LCD [28]. The light is first collected by an imaging lens, which images the FOV of the sensor onto an LCD positioned at the image plane. When an LCD pixel is set to black, no light is transmitted, whereas when a pixel is set to white, light is transmitted. The pattern of black and white pixels on the LCD combined with the imaging lens enables the limitation of the FOV of the sensor to an arbitrary shape. The light exiting the LCD is then incident on a diffuser, which ensures that the light is at least partially directed toward the AS7341 multispectral sensor. The sensor concept is illustrated in Fig. 1.

The four components are described in Sections II-A–II-D, and equations characterizing the transmission of the light are provided.

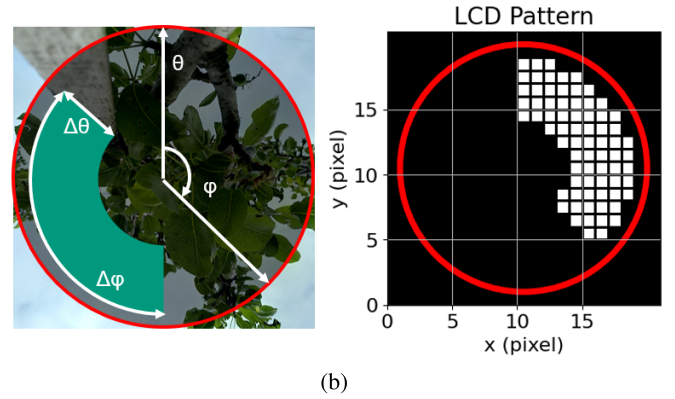
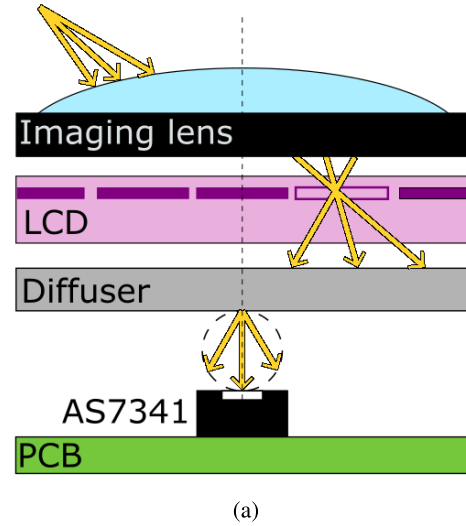


Fig. 1. Illustration of the novel sensor concept. (a) Components of the system and path of the incident light. (b) Example LCD pattern corresponding to the FOV in green.

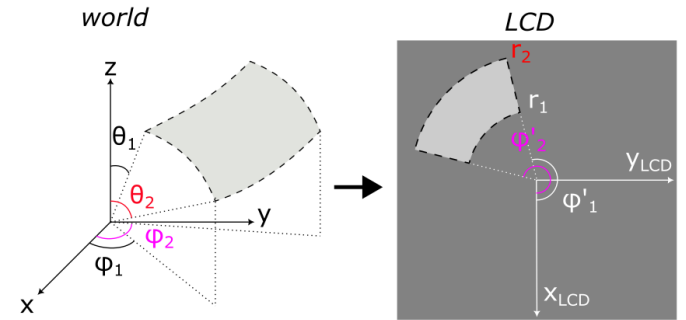


Fig. 2. LCD mapping illustration. The FOV defined by the ranges (θ_1, θ_2) and (ϕ_1, ϕ_2) in the world is mapped to the ranges (r_1, r_2) , and (ϕ'_1, ϕ'_2) on the LCD.

A. Imaging Lens

The first element of the sensor is the imaging lens, which images objects present in the world. The world coordinate system is defined in spherical coordinates and the image plane is defined in polar coordinates, with the origin being situated in both cases at the optical axis of the lens. The light originating from a direction (θ, ϕ) in the world is thus imaged on a point (r, ϕ') on the image plane (Fig. 2).

The mapping of θ to r is done through the lens projection formula, which can be expressed generally as [29]

$$r(\theta) = k_1\theta + k_2\theta^3 + k_3\theta^5 + \dots \quad (1)$$

where $k_1, k_2, k_3, \dots$ are parameters specific to each lens, with θ in rad and r in pixels.

The mapping of ϕ to ϕ' is defined by the following offset, because of the image inverting property of lenses:

$$\phi' = \phi + \pi \quad (2)$$

with ϕ, ϕ' in rad.

Aside from the world-to-image mapping, it is also important to quantify the transmittance of the lens, which depends on the wavelength and the incident direction of the light. The latter is due to vignetting: a decrease in light transmission as θ increases [30]. In the present case, natural vignetting, also called the “ $\cos^4\theta$ law of illumination,” is the predominant vignetting effect. To improve or nullify the impact of natural vignetting on wide-angle lenses, a graduated neutral density filter, or spot filter, can be used. Its density is maximal at $\theta = 0^\circ$ and near zero at higher θ values, such that the θ dependence of the transmittance is quasi-nullified, at the price of an overall lower transmittance [30].

Lenses are also impacted by changes in temperatures in two ways: the lens geometry deforms due to thermal expansion and the refractive index changes [31]. While the former is hard to predict in practice, the latter is well-known and causes a change in the focal length of a lens. This effect, termed the thermal focal shift [30], is most important for plastic lenses because of their higher thermo-optic coefficient of change of refractive index. Plastics exhibit a thermo-optic coefficient of approximately -0.003 K^{-1} [30], which is three orders of magnitude higher than typical optical glass materials [32]. For this reason, a glass lens should be used.

Formally, the incident spectral irradiance $I_{\lambda,sky}(\theta, \phi)$ ($\text{W m}^{-2} \text{ nm}^{-1}$) is captured by the lens and imaged at position (r, ϕ') on the image plane. The resulting spectral irradiance $I_{\lambda,img}(r, \phi')$ ($\text{W m}^{-2} \text{ nm}^{-1}$) is given by

$$I_{\lambda,img}(r, \phi') = \alpha_\lambda(T) \xi(\theta, \phi) I_{\lambda,sky}(\theta, \phi) \quad (3)$$

where $\alpha_\lambda(T)$ is the lens spectral transmittance (assumed to be temperature-dependent) and $\xi(\theta, \phi)$ is the spatially dependent vignetting. Through the projection formula and the inversion property of imaging lenses, coordinates can be transformed between (θ, ϕ) and (r, ϕ') .

B. LCD

The second component in the path of light is the LCD. The LCD acts as an optical shutter to transmit or block light from specific areas of the FOV. LCDs rely on the anisotropic properties of liquid crystals, a fluid composed of long rod-shaped molecules. Applying an electric field across a liquid crystal modifies its anisotropy, enabling the controlled modification of the polarization of incident light [17]. A simple LCD is composed of a liquid crystal cavity sandwiched between two perpendicular linear polarizers [33], such that the variation of the polarization leads to a variation of the transmittance of

light across the cell. ON and OFF portions of the LCD have a spectral transmittance of $\beta_{\lambda,ON}(T)$ and $\beta_{\lambda,OFF}(T)$, respectively. The LCD contrast $\sigma_\lambda(T) = (\beta_{\lambda,ON}(T)/\beta_{\lambda,OFF}(T))$ should be kept high, to ensure that only the light from the desired FOV is transmitted.

The spectral irradiance coming out of the LCD $I_{\lambda,LCD}(r, \phi')$ ($\text{W m}^{-2} \text{ nm}^{-1}$) can be expressed as

$$I_{\lambda,LCD}(r, \phi') = \begin{cases} I_{\lambda,img}(r, \phi') \beta_{\lambda,OFF}(T) \sigma_\lambda(T), & \forall (r, \phi') \in D_{ON} \\ I_{\lambda,img}(r, \phi') \beta_{\lambda,OFF}(T), & \forall (r, \phi') \in D_{OFF} \end{cases} \quad (4)$$

where D_{ON} and D_{OFF} are the regions on the image plane where the LCD is transparent and opaque, respectively.

C. Diffuser

The third component in the path of the light is the diffuser, which ensures that light transmitted through any point on the LCD is scattered toward the AS7341 light sensor. The AS7341 manufacturer recommends the use of a bulk diffuser with scattering characteristics approaching a cosine response [34]. The temperature-dependent spectral transmittance of the diffuser is expressed by $\psi_\lambda(T)$. Additionally, the amount of light diffused toward the AS7341 (situated at the optical axis) varies depending on the point of incident of incoming rays onto the diffuser. This effect is expressed by the spatially-dependent spectral transmittance $\rho_\lambda(r, \phi')$. The spectral irradiance coming out of the diffuser $I_{\lambda,diff}$ ($\text{W m}^{-2} \text{ nm}^{-1}$) onto the light sensor is thus the integral of the spectral irradiance coming from every position on the LCD $I_{\lambda,LCD}(r, \phi')$ multiplied by the infinitesimal area element dA over the total region D

$$I_{\lambda,diff} = \frac{\psi_\lambda(T)}{A_D} \iint_D \rho_\lambda(r, \phi') \cdot I_{\lambda,LCD}(r, \phi') dA \quad (5)$$

where $A_D = \iint_D dA$ is the area of the total LCD region D .

D. AS7341 Light Sensor

The AS7341 multispectral sensor is a light sensor consisting of an array of photodiodes, each responsive to a different range of the light spectrum. Table I lists the characteristics of each channel.

The photocurrent generated by light incident on one of the photodiodes is converted to a 16-bit digital value via the six ADC channels. The raw ADC count raw_{ch} is related to the incident irradiation via the equation

$$\frac{raw_{ch}}{t_{int} \cdot gain} = basic_{ch} = \int_0^\infty \tau_{ch}(\lambda) I_{\lambda,diff} d\lambda \quad (6)$$

where t_{int} is the total integration time (ms), gain is the ADC gain (from $0.5 \times$ to $512 \times$), $basic_{ch}$ is the unitless sensor value independent of the ADC settings, and τ_{ch} is the irradiance responsivity for a given channel ($\text{m}^2 \text{ W}^{-1}$).

The AS7341 contains six ADC channels for eight visible channels, one clear, and one NIR channel (for a total of ten channels). This implies that two cycles are necessary to sample all the channels: in the first cycle, the first half of the

TABLE I
AS7341 OPTICAL CHANNELS CHARACTERISTICS.
ADAPTED FROM [34]

Channel	Center Wavelength (nm)	FWHM ¹ (nm)
F1	415	26
F2	445	30
F3	480	36
F4	515	39
F5	555	39
F6	590	40
F7	630	50
F8	680	52
NIR	910	n/a
Clear	Si response (no filter)	n/a

¹Full width at half maximum.

channels are assigned to the ADC, and in the second cycle, the second half of the channels are assigned to the ADC. The total measurement time is therefore $2t_{\text{int}}$.

Since the raw_{ch} values are the output of an ADC, they exhibit a quantization error ϵ_{quant} corresponding to $\pm 1/2$ least significant bit (LSB). To keep the relative error minimal, the gain and integration time should be set to ensure high raw_{ch} , ideally close to the ADC full scale.

The temperature dependence of the photodiodes is compensated internally using a reference dark photodiode [34].

The advantage of using a spectral sensor with several narrowband channels compared to a single broadband Si photodiode is that the sensor can be calibrated to different definitions of PAR (see Section III), without hardware changes by weighting the channels differently. Further, it enables us to obtain additional information about the spectral composition of the light.

III. PAR MEASUREMENT MODEL

PAR is the amount of light that can be used by a plant during photosynthesis, defined as the photon flux per ground area per second ($\mu\text{mol m}^{-2}\text{s}^{-1}$). Plants absorb light in the range of 340–760 nm and this absorption varies with the wavelength λ (nm). This λ -dependent use of incident light for photosynthesis is called action spectrum [35].

There are two approaches to measuring PAR [3]; the first consisting in weighing the photons based on their wavelength to reflect the action spectrum. This is also called the yield photon flux (YPF) and is defined by

$$\text{PAR}_{\text{YPF}} = \frac{10^6}{hcN_A} \int_{360\text{ nm}}^{760\text{ nm}} \omega(\lambda) \cdot \lambda I_{\lambda}(\lambda) d\lambda \quad (7)$$

where h is Planck's constant (Js), c is the speed of light (ms^{-1}), N_A is Avogadro's number (mol^{-1}), I_{λ} is the spectral irradiance density (irradiance per unit wavelength, in $\text{W m}^{-2} \text{nm}^{-1}$), and $\omega(\lambda)$ is the normalized average action spectrum of plants.

A second approach consists in ignoring the action spectrum, and instead assuming that incident light acts in photosynthesis irrespective of its wavelength. This is also called the

photosynthetic photon flux (PPF) and is defined by

$$\text{PAR}_{\text{PPF}} = \frac{10^6}{hcN_A} \int_{400\text{ nm}}^{700\text{ nm}} \lambda I_{\lambda}(\lambda) d\lambda. \quad (8)$$

Note that PAR_{PPF} is calculated over a smaller range of λ . In this work, the PAR_{PPF} is used and, unless mentioned otherwise, the term PAR refers to PAR_{PPF} .

To measure the PAR, $I_{\lambda}(\lambda)$ must be expressed in terms of basic_{ch} values measured by the AS7341 sensor. In the first step, the inverse relationship is expressed, namely basic_{ch} in terms of $I_{\lambda, \text{sky}}(\theta, \phi)$. This is realized by combining (3)–(6) and performing the coordinates transformation $(r, \phi') \rightarrow (\theta, \phi)$ with (1)–(2). The derivation is shown in the Appendix and results in

$$\begin{aligned} \text{basic}_{ch} = & \frac{1}{A_D} \int_0^{\infty} S_{\lambda} \tau_{ch}(\lambda) \psi_{\lambda}(T) \beta_{\lambda, \text{OFF}}(T) \alpha_{\lambda}(T) \\ & \cdot \left(1 + \frac{\sigma_{\lambda}(T) - 1}{S_{\lambda}} \cdot \iint_{D_{\text{ON}}} g_{\lambda}(\theta, \phi) d\theta d\phi \right) d\lambda \end{aligned} \quad (9)$$

where $g_{\lambda}(\theta, \phi) = \rho_{\lambda}(\theta, \phi) \xi(\theta, \phi) I_{\lambda, \text{sky}}(\theta, \phi) r(\theta) (dr(\theta) / d\theta)$ is a positive function and $S_{\lambda} = \iint_D g_{\lambda}(\theta, \phi) d\theta d\phi$ is a constant.

Equation (9) describes the value measured by each channel of the AS7341 sensor for a given incident irradiation $I_{\lambda, \text{sky}}$. However, to calculate the PAR, an expression for $I_{\lambda, \text{sky}}$ in terms of basic_{ch} is required. This expression cannot be directly obtained from (9) because of the three integrals over θ , ϕ , and λ . To proceed further, simplifications are required.

- 1) The AS7341 yields measurements in ten discrete channels. The PAR can therefore be approximated by a linear sum of the contributions of each channel [36]

$$\text{PAR} \approx \sum_{ch} \omega_{ch} I_{ch, \text{sky}} \quad (10)$$

where $I_{ch, \text{sky}}$ is the incident irradiance corresponding to each channel (W m^{-2}) and ω_{ch} are the per-channel weights. This enables the transformation of the spectral quantities to discrete values (one for each of the ten channels of the sensor), which eliminates the integral over λ . The discrete values are denoted by the subscript ch

$$\begin{aligned} \int I_{\lambda} d\lambda \left(\text{W m}^{-2} \text{nm}^{-1} \right) & \rightarrow I_{ch} \left(\text{W m}^{-2} \right), \text{ etc.} \\ \rho_{\lambda}(\theta, \phi) & \rightarrow \rho_{ch}(\theta, \phi), \text{ etc.} \end{aligned}$$

- 2) $I_{ch, \text{sky}}$ is assumed to be constant across the sky, such that

$$I_{ch, \text{sky}}(\theta, \phi) = I_{ch, \text{sky}} \quad \forall \theta, \phi.$$

This enables factoring out $I_{ch, \text{sky}}$ from the double integral on θ and ϕ . Note that this assumption does not hold in practice under direct sunlight, as the sky irradiance has a sharp maximum at the sun position. Under indirect sunlight (heavily overcast sky), the sky irradiance spatial distribution flattens, but a peak is still visible at a zenith angle of 0° [37].

- 3) Since $g_{ch}(\theta, \phi)$ is a positive function and D_{ON} is a subset of D , the double integral term can be replaced by an arbitrary function $\delta(\theta_1, \theta_2, \phi_1, \phi_2)$

$$\begin{aligned} \delta_{ch}(\theta_1, \theta_2, \phi_1, \phi_2) &= \frac{1}{S_{ch}} \cdot \iint_{D_{ON}} g_{ch}(\theta, \phi) d\theta d\phi \\ &= \int_{\theta_1}^{\theta_2} \gamma_{ch,\theta}(\theta) d\theta \cdot \int_{\phi_1}^{\phi_2} \gamma_{ch,\phi}(\phi) d\phi \end{aligned} \quad (11)$$

where the region D_{ON} is bounded by the conditions $\theta \in [\theta_1, \theta_2]$ and $\phi \in [\phi_1, \phi_2]$, and $\gamma_{ch,\theta}(\theta)$ and $\gamma_{ch,\phi}(\phi)$ are arbitrary functions, defined such that the following conditions are fulfilled:

$$\begin{aligned} 0 &\leq \int_{\theta_1}^{\theta_2} \gamma_{ch,\theta}(\theta) d\theta \leq 1 \quad \forall \theta \in [0^\circ, \theta_{max}] \\ 0 &\leq \int_{\phi_1}^{\phi_2} \gamma_{ch,\phi}(\phi) d\phi \leq 1 \quad \forall \phi \in [0, 2\pi] \\ 1 &= \int_{0^\circ}^{\theta_{max}} \gamma_{ch,\theta}(\theta) d\theta \\ 1 &= \int_{0^\circ}^{2\pi} \gamma_{ch,\phi}(\phi) d\phi. \end{aligned} \quad (12)$$

The channel-specific parameters can then be combined into a new temperature-dependent linear regression parameter $\omega'_{ch}(T)$

$$\omega'_{ch}(T) = \frac{\omega_{ch} A_D}{\tau_{ch}(T) \psi_{ch}(T) \alpha_{ch}(T) \beta_{ch,OFF}(T) S_{ch}}. \quad (13)$$

Those simplifications are then applied to (9), which allows isolating I_{ch} and inserting it into (10). This yields our simplified PAR model

$$PAR \approx \sum_{ch} \frac{\omega'_{ch}(T) \cdot basic_{ch}}{\delta_{ch}(\theta_1, \theta_2, \phi_1, \phi_2) \cdot (\sigma_{ch}(T) - 1) + 1}. \quad (14)$$

When the FOV is restricted, less light is incident onto the sensor, such that $basic_{ch}$ diminishes. This is compensated by the δ_{ch} term, which also diminishes as the FOV is decreased. The terms $\delta_{ch}(\sigma_{ch} - 1)$ and 1 in the denominator represent, respectively, the contributions of light incident through the D_{ON} and D_{OFF} regions of the LCD. An ideal sensing system would only measure light incident from the D_{ON} region. To tend toward that ideal case in practice, the following condition must be satisfied:

$$\delta_{ch}(\sigma_{ch} - 1) \gg 1. \quad (15)$$

IV. MATERIALS AND METHODS

Having previously described the sensor concept and derived a simplified model enabling the calculation of the PAR from the AS7341 sensor values, this section describes the realization, calibration, and testing of a sensor prototype.

A. Sensor Prototype

A sensor prototype is built to evaluate the feasibility of the sensor concept. At this stage, achieving the lowest power consumption is not a goal. The selection of the components is instead mostly guided by insights from the PAR model (14).

- 1) The transmission coefficients of each component are multiplied [see (13) and (11)], which leads to a rapid decrease in overall transmission through the system if one or more of the components exhibits a low transmission. Components should be selected with as high a transmission coefficient as possible.
- 2) The LCD Contrast σ_{ch} should be as high as possible to satisfy the condition dictated by (15).

For the imaging lens, the 140° LN056 M12 lens from Arducam (Kowloon, Hong Kong) is chosen. It is preferred to a wider-angled lens (such as 180°) for two reasons. First, the amount of light incident from $\theta > 70^\circ$ is relatively small compared to the light incoming from $\theta < 70^\circ$. Second, the attenuation due to natural vignetting is lower at 70° , so the compensation filter transmits more light than for a wider angle lens.

The selection of the LCD is guided by the transmittance and the contrast, which should both be as high as possible. The ideal LCD would therefore be a segment LCD. The individual control of the actuation signal of each segment enables theoretically infinite contrast, and the simple stack of glass, electrodes, and polarizers yields a maximal transmission close to 50%. Custom segment LCDs can be ordered, but prototyping costs are hefty and production runs have a high minimum quantity. Instead, it is decided to use a commercial matrix LCD, which can display any desired pattern at the cost of a higher power consumption.

Commercial LCDs are rarely sold transparent, but some models can be modified by taking out the backlight or the back reflector, which makes them transparent. Two LCDs from Display Visions (Gilching, Germany) are considered: a passive matrix and an active matrix. The passive matrix is the EA DOGL128S-6, a black and white 128×64 pixels display with a reflector and a pixel size of $500 \times 460 \mu\text{m}^2$. No backlight is required, as for this application the light incident on the display must be transmitted toward the optical sensor and not reflected back. The active matrix LCD is the EA TFT015-22AI, a backlit 1.5" 240×240 pixels TFT LCD. Each pixel, of total area $115.5 \times 115.5 \mu\text{m}^2$, is composed of three red, green, and blue parts, such that they can be made to appear in any color. The main differences between both devices is that the EA DOGL128S-6 passive matrix LCD has a higher transmittance, whereas the EA TFT015-22AI active matrix has a higher contrast and power consumption. The latter is selected, because the high contrast is important to enable the restriction to small FOVs [see condition in (15)], and the power consumption is not critical at the prototype stage. It is made transparent by removing the backlight.

Two diffusers are considered. The first one is 3 mm-thick PMMA, which acts as a volume diffuser and possesses a high resistance to UV light [38]. The other diffuser considered is the diffuser included with the AS7341 Evaluation Kit, the Kimoto 100 PBU (Kimoto Tech, Georgia, USA) surface diffuser. The PMMA is chosen for this work as it has high UV resistance and it has been used in a previous AS7341-based sensor node [26].

The sensor prototype also includes a digital temperature sensor (MCP9808, Microchip Technology, Arizona, USA) to

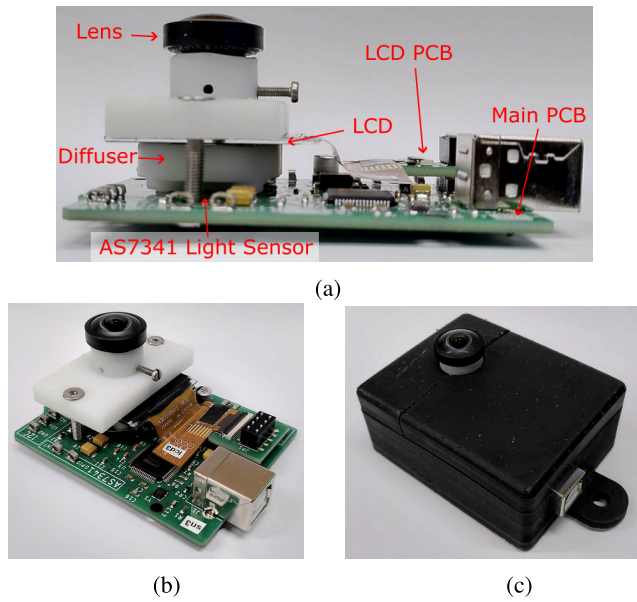


Fig. 3. Pictures of an assembled sensor node. (a) Without housing (side view). (b) Without housing. (c) With housing.

account for the temperature dependence of the terms of the PAR Model (14) of the above components. To facilitate testing and characterization, a USB port is used both for communication and power. A USB to dual serial bridge (FT2232H, FTDI, Glasgow, Scotland) enables direct communication over USB to the sensor node components via I²C and SPI channels.

Two low-dropout voltage regulators are used to obtain required supply voltages from the USB 5 V: LF33 (STMicroelectronics, Geneva, Switzerland) for 3.3 V and MCP1801 (Microchip, Arizona, USA) for 1.8 V.

The electronic components are assembled on two two-layer printed circuit boards (PCBs). Two PCBs are required to account for the fact that the LCD has to lay flat over the diffuser covering the light sensor. With this geometry, a direct connection of the LCD on the main PCB would create risks of strain and rupture in the short flex cable. A second PCB is therefore included for the sole purpose of connecting with the LCD.

Three sets of mechanical parts are designed and 3D printed with a low-force stereolithography printer (Form 3+, Formlabs, Boston, USA).

The assembled sensor prototype is shown in Fig. 3. The total cost of materials for one sensor prototype is 117 USD.

The sensor prototype is estimated to require 292.14 mW in the measurement state and 6.331 mW in a standby state (see Table II). Since the USB-to-serial bridge has a high-power consumption and would not be part of an actual low-power sensor, the total power has also been estimated without this component. A more realistic estimate is therefore a consumption of 67.14 mW in the measurement state and 5.398 mW in a standby state.

B. LED Test Sphere

To evaluate the sensor's ability to distinguish light coming from a specific direction (θ , ϕ), an LED test sphere was

TABLE II
ESTIMATED SENSOR PROTOTYPE POWER CONSUMPTION

Component	Standby		Measurement	
	Power (mW)	%Total (%)	Power (mW)	%Total (%)
Light sensor (AS7341)	0.009 ¹	0.14	0.54 ¹	0.18
LCD (EA TFT015-22AI)	0.132 ¹	2.09	24.78 ¹	8.48
Temperature sensor (MCP9808)	0.007 ¹	0.10	1.32 ¹	0.45
USB-to-serial bridge (FT2232H)	0.933 ^{2,3}	14.74	225 ²	77.02
5V to 3.3V (LF33CDT)	5.000 ¹	78.98	40 ¹	13.69
5V to 1.8V (MCP1801T)	0.250 ¹	3.95	0.50 ¹	0.17
Total	6.331	100	292.14	100
Total (without FT2232H)	5.398	85.25	67.14	22.98

¹Maximal specification from datasheet.

²Typical specification from datasheet.

³USB Suspend state.

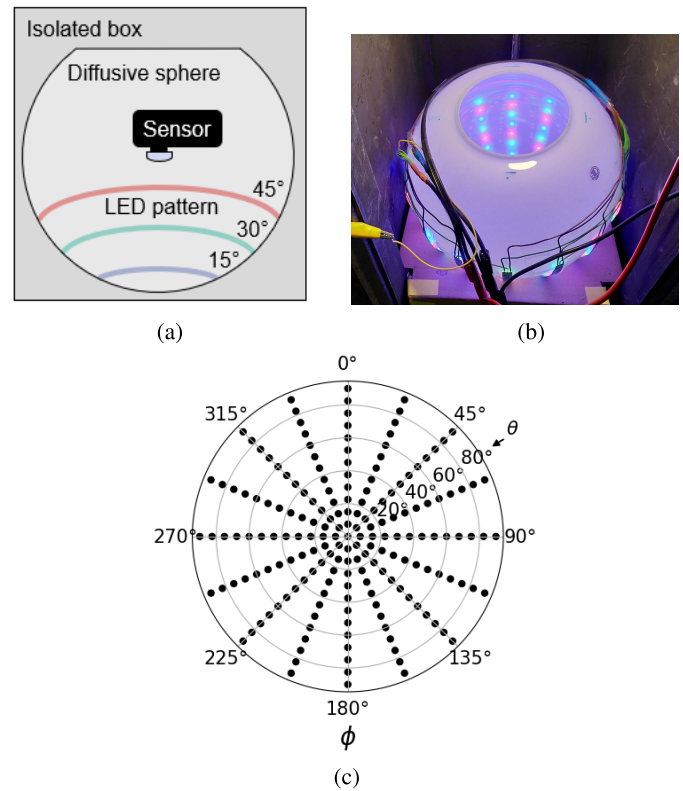


Fig. 4. Measurement setup consisting of an LED test sphere with individually addressable RGB LEDs. The sensor is inserted in the hole. (a) Diagram. (b) Picture. (c) LED positions inside the sphere.

built using 185 individually addressable RGB LEDs (SK6812) placed on the outside of a diffusing sphere (24 cm Fado Table Lamp, IKEA, Sweden), facing inward. The sensor is then placed inside, looking at the LEDs. Fig. 4 shows a picture of the measurement setup, a diagram of the sensor's position inside the sphere, and a map of the location of the LEDs inside the sphere. The error in position for the LEDs is estimated to be $\pm 2^\circ$ in the θ -direction and $\pm 6^\circ$ in the ϕ -direction.

This test setup, while offering the possibility to control the color and direction of incoming light, differs from natural sunlight in two important factors. First, the maximum illumination inside the sphere corresponds to a PAR of $200 \mu\text{mol m}^{-2}\text{s}^{-1}$.

On the other hand, PAR under clear skies (direct sunlight) has average values of $541\text{--}1753 \mu\text{mol m}^{-2}\text{s}^{-1}$ depending on the sun zenith angle, while an obscured sky (diffuse sunlight) has an average PAR of $509 \mu\text{mol m}^{-2}\text{s}^{-1}$ [39]. Second, the light spectrum inside the sphere does not resemble sun illumination. It is rather composed of three peaks, due to the individual red, green, and blue LEDs, which correspond, respectively, to the $F3$ (630 nm), $F4$ (515 nm), and $F7$ (480 nm) channels of the AS7341.

C. Calibration Methods

This section introduces the procedures required to go from an assembled sensor prototype to a prototype that yields accurate and valid PAR measurements. The first step concerns the mechanical alignment of the lens: it must be set so that the image plane lies in the middle of the LCD device. Then, the lens projection formula is measured. Finally, the parameters of the PAR model (14) are determined experimentally, namely the LCD contrast $\sigma_{ch}(T)$, the delta function $\delta_{ch}(\theta_1, \theta_2, \phi_1, \phi_2)$, and the combined linear regression parameters ω'_{ch} .

1) **Lens Image Plane Position:** To ensure that the sensor works properly, the lens must be adjusted so that the image plane position z_{img} lies approximately in the middle of the LCD's liquid crystal cavity. For that purpose, a C270 HD webcam (Logitech, Lausanne, Switzerland) is used, and the procedure is illustrated in Fig. 5.

The webcam lens is taken out so that the image sensor is exposed. The sensor node lens is placed in the lens holder, and the lens holder is held flat against the image sensor [see Fig. 5(a)]. The webcam output is viewed on the computer, and the lens position is adjusted so that the image becomes in focus. At this moment, the image plane is directly on the image sensor. However, the image sensor surface sits slightly over the PCB (the bottom surface of the lens holder), whereas the center of the LCD sits slightly below the bottom surface of the lens holder [see Fig. 5(b)]. The image plane must therefore be lowered by 0.5 mm. The threading of the lens is 0.5 mm, so a full turn corresponds to a vertical movement of the image plane $\Delta z_{\text{img}} = -0.5 \text{ mm}$.

2) **Lens Projection Formula:** The projection formula $r(\theta)$ of the lens can be described with the general projection formula (1). The goal here is thus to determine the parameters k_i .

To determine the projection formula, it is first required to know the position of the lens optical axis onto the LCD, i.e., the pixel on the LCD that corresponds to an object at $\theta = 0^\circ$ in the world. This ensures that the patterns used to restrict the FOV to a specific area are aligned with the lens's optical axis. The sensor prototype is placed in the LED test sphere, and the single LED at $\theta = 0^\circ$ is lit up with the color white and maximum brightness. Each pixel of the LCD is individually toggled ON and OFF one after the other. The pixel for which $\text{basic}_{\text{clear}}$ is maximal corresponds to the position of the lens's optical axis.

The projection formula can then be experimentally determined, also using the LED test sphere. All LEDs placed at a certain position θ are lit up white at maximum brightness. Then, 1 pixel-thick circular rings are turned ON on the LCD, each centered at the center pixel determined previously [see

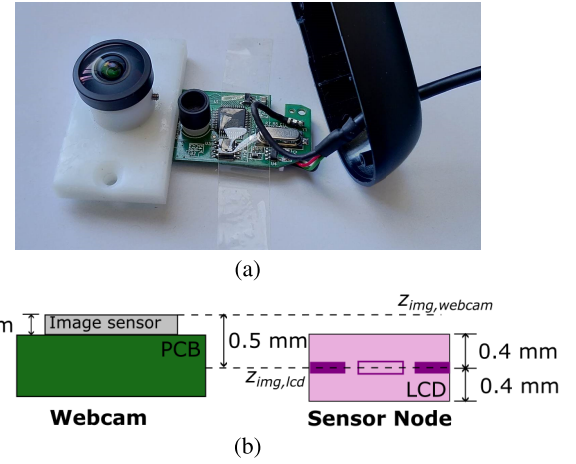


Fig. 5. Measurement procedure to set the proper lens image plane position. (a) Lens holder is pressed against the exposed image sensor of a Logitech C270 HD webcam, the lens is screwed in so that the image goes from being out of focus to in focus. (b) Lens is screwed in a full turn to account for the Difference in position between the image sensor $z_{\text{img},\text{webcam}}$ and the center of the LCD $z_{\text{img},\text{lcd}}$.

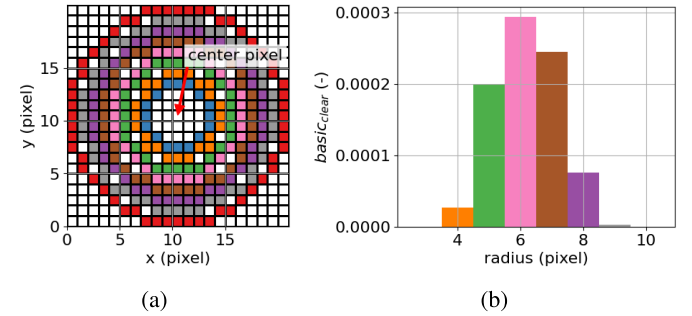


Fig. 6. Illustration of the procedure used to measure the projection formula. In this example, all LEDs at $\theta = 15^\circ$ are lit up, and (a) 1 pixel-thick circular patterns of increasing radius are displayed on the LCD. Only one ring is displayed at a given time. For each circular ring, (b) measurement is made. The radius for which the irradiation is maximal (here $r = 6$ pixels) thus corresponds to $\theta = 15^\circ$.

Fig. 6(a)]. For each ring, the light is measured with the sensor. Thus, for a given circle of LEDs at θ , a distribution of the irradiation in the clear channel ($\text{basic}_{\text{clear}}$) in terms of the radius can be obtained [see example in Fig. 6(b)]. The radius for which $\text{basic}_{\text{clear}}$ is maximal is chosen as the radius corresponding to that θ . This procedure is repeated for all LEDs in the test sphere up to $\theta = 52.5^\circ$. LEDs positioned at $\theta > 52.5^\circ$ are not used because the attenuation due to their high θ value is such that there is not enough light traversing 1 pixel-thick circular rings.

3) **LCD Contrast:** σ_{ch} can be expressed from two measurements of the sensor, one with all pixels turned ON and one with all pixels turned OFF

$$\sigma_{ch} = \frac{\beta_{ch,\text{ON}}}{\beta_{ch,\text{OFF}}} = \frac{I_{ch,\text{ON,out}}/I_{ch,\text{in}}}{I_{ch,\text{OFF,out}}/I_{ch,\text{in}}} = \frac{\text{basic}_{ch,\text{ON}}}{\text{basic}_{ch,\text{OFF}}}. \quad (16)$$

To account for the temperature dependence, the measurements are conducted in a climate chamber (LHU-113 Temperature and Humidity Cabinet, ESPEC North America, Michigan, USA). Since little light is transmitted through the LCD when all pixels are OFF (opaque), the measurements are

made without the lens to maximize overall transmission. The sensor node without a lens is placed in the climate chamber under a white LED light bulb. The temperature of the climate chamber is swept down from 45 °C to 0 °C in steps of 5 °C. σ_{ch} is measured when the temperature is stable.

The temperature dependence of $\sigma_{ch}(T)$ is found experimentally to be best modeled by a second-order polynomial, such that

$$\sigma_{ch}(T) \approx k_1 + k_2 \cdot T + k_3 \cdot T^2 \quad (17)$$

where k_1 , k_2 , and k_3 are linear regression coefficients.

4) **Delta Functions:** The function $\delta_{ch}(\theta_1, \theta_2, \phi_1, \phi_2)$ is composed of the two functions $\gamma_{ch,\theta}$ and $\gamma_{ch,\phi}$, which both must be measured experimentally. In both cases, the measurements are conducted in the LED test sphere with a sensor node from which the LCD has been taken out. The LCD is taken out because δ_{ch} does not depend on LCD parameters, but rather only on the imaging lens and the diffuser [see (11)].

For the determination of $\gamma_{ch,\theta}$, LEDs positioned in a concentric ring at θ are lit up, and the irradiance is measured. Measurements are taken for concentric rings from $\theta = 7.5^\circ$ to $\theta = 67.5^\circ$. Then, each irradiance measurement is divided by the total number of LEDs n_{LEDs} for that position. This is done to account for the fact that there are eight LEDs for $\theta = 7.5^\circ$ and 16 for $\theta > 15^\circ$ [see map of LED positions in Fig. 4(c)]. The measured data points are then fit to a linear sum of arbitrary functions, chosen to best represent the measured data. Finally, this fit is normalized over the whole range $\theta \in [0^\circ, 70^\circ]$.

For the determination of $\gamma_{ch,\phi}$, the same procedure is followed for the ϕ -direction. LEDs positioned in a line at the same ϕ are lit up, and the irradiance is measured. Measurements are taken for lines from 0° to 337.5° . To ensure that n_{LEDs} is constant, only LEDs in the range $\theta \in [15^\circ, 67.5^\circ]$ are lit up. The measured data points are then fit to a linear sum of arbitrary functions, chosen to best represent the measured data. Finally, this fit is normalized over the whole range $\phi \in [0^\circ, 360^\circ]$.

5) **PAR Model Regression:** To convert the eight channels of the AS7341 sensor into one PAR value, a multiple regression method is needed. Due to the overlapping of the channels (see Table I), the measured $basic_{ch}$ values are correlated among channels, which means if one channel changes, another channel changes similarly. A common linear regression method suffers under collinearity and is therefore unsuitable to determine the regression parameters ω'_{ch} [40]. A good alternative, working well with highly correlated data, is Partial Least Squares (PLS) [41]. PLS is a linear regression method that converts the nine input channels into so-called latent variables (LVs). The LVs correspond to linear combinations of the input matrix and have two effects: minimizing the cross correlation of the spectral channels and maximizing the correlation with the responses. This principle is often used for spectral measurements in biology or chemistry [42], [43]. The number of LVs of a PLS can go from one up to the rank of the input matrix (the number of linearly independent channels). The ideal number of LVs is determined either by the smallest residual error or by a predefined threshold.

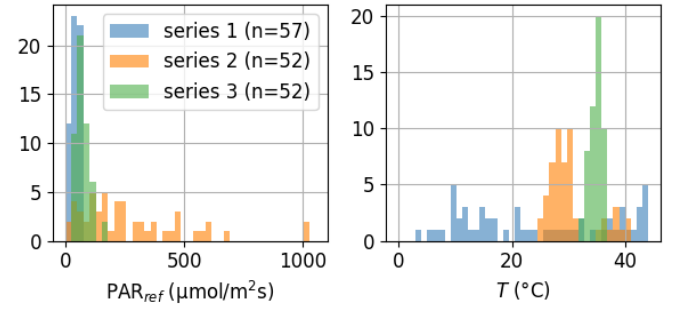


Fig. 7. Histograms of the reference PAR PAR_{ref} and temperature T spread for the training data. The series contain a total of 161 data points.

Since the underlying model of PLS assumes a linear model, we have to introduce the polynomial regression approach to compensate for the non-linear behavior of the system due to temperature dependence [41]. The temperature dependence of $\omega'_{ch}(T)$ is experimentally measured to be mainly quadratic, such that

$$\omega'_{ch}(T) \approx h_{ch,0} + h_{ch,1} \cdot T^2 \quad (18)$$

where $h_{ch,0}$ and $h_{ch,1}$ are fit parameters and T is the temperature (°C).

The model input therefore contains two fit parameters for every channel. The considered channels are F1–F8 and clear, which totals to 18 fit parameters.

The data used to perform the PLS is composed of three data series. Each series contains data measured with the sensor node in combination with reference PAR values obtained with the LI-190R Quantum Sensor.

The three data series are as follows.

- 1) temperature sweeps under different illumination conditions in a climate chamber. Transparent papers of different colors are placed between the light source (LED 8.5 W/2700 K, Osram GmbH, Munich, Germany) and the sensor to simulate a variation in the incident light spectrum. This series contains 57 data points across a wide temperature range from 4 to 44 °C.
- 2) measurement under diffused sunlight with transparent papers of different colors placed between the sun and the sensor. This series contains 52 data points across a wide PAR range taken on the campus of the faculty of engineering of the University of Freiburg.
- 3) measurements realized inside the LED test sphere with different combinations of brightnesses for the red, green, and blue LEDs. This series contains 52 data points across a narrow PAR range.

The repartition of PAR values and temperature in the three data series is shown in Fig. 7.

The PLS regression is implemented in Python with the PLSRegression class from scikit-learn 1.3.0 [44].

The calibration data is split in 75% training set ($n_{\text{training}} = 121$) and 25% test set ($n_{\text{test}} = 40$) with random state = 42. A tenfold cross-validation is used to evaluate the optimal number of LVs for the smallest normalized root mean squared error (nRMSE) and mean absolute percentage error (MAPE) of the training set. These metrics are defined in Appendix B. The

coefficients of the PLS model then correspond to the variables $h_{ch,0}$ and $h_{ch,1}$ (18) of regression parameters ω'_{ch} . For easier calculations the PLSRegression algorithm mean-centers and scales the training set before applying the regression.

The performance of the regression is later evaluated on the test set using the r^2 coefficient, as well as the mean absolute error (MAE), MAPE, root-mean-square error (RMSE), and nRMSE. These metrics are also defined in Appendix B.

D. Characterization Methods

Following the calibration of the sensor prototype, experiments are conducted in the lab to characterize it. First, the LED test sphere is used to measure the system's resolution. Then, the power consumption of the prototype is measured.

1) **System Spatial Resolution:** The spatial resolution of an optical system can be obtained by measuring the width of the line spread function (LSF) at half of its maximum [45]. This distance, also called full-width at half maximum (FWHM), represents the minimum distance between two objects that can be distinguished by the system.

To evaluate the resolution of the system in the θ -direction, all LEDs at $\theta_{LED} = 7.5^\circ$ are lit up white. Then, the FOV is restricted to a thin “donut” shape: the ϕ range $[0^\circ, 360^\circ]$ is constant and the θ range is swept from $[0^\circ, 2.5^\circ]$ to $[67.5^\circ, 70^\circ]$, in increments of 2.5° . The measured $basic_{clear}(\theta)$ represents the LSF of the system in the θ -direction at the position of θ_{LED} . To measure the $FWHM_\theta$, the LSF is fit to a Gaussian curve with the equation

$$LSF(\theta) = basic_{clear}(\theta) = A_\theta \exp\left(-\frac{(\theta - M_\theta)^2}{2V_\theta}\right) \quad (19)$$

where A_θ is a scaling factor, M_θ is the mean, and V_θ is the variance. The FWHM is then approximated by

$$FWHM_\theta = 2\sqrt{2V_\theta \ln 2} \approx 2.355\sqrt{V_\theta}. \quad (20)$$

These steps are repeated for LEDs positioned at θ_{LED} up to 52.5° .

To evaluate the resolution of the system in the ϕ -direction, an equivalent procedure is followed. All LEDs at $\phi_{LED} = 0^\circ$ are lit up white. Then, the FOV of the sensor is restricted to a thin “pie slice” shape: the θ range $[0^\circ, 70^\circ]$ is constant and the ϕ range is swept from $[0^\circ, 5^\circ]$ to $[355^\circ, 360^\circ]$, in increments of 5° . The measured $basic_{clear}(\phi)$ represents the LSF of the system in the ϕ -direction at the position of ϕ_{LED} . Like for $FWHM_\theta$, $FWHM_\phi$ is calculated by fitting the measured values to a Gaussian curve [see (19) and (20)]. These steps are repeated for LEDs positioned at ϕ_{LED} up to 337.5° .

2) **Power Consumption:** The power consumption is measured both for the sensor prototype as a whole and for the LCD individually. The sensor prototype power is measured with the CT-3 USB Power Meter (AVHzY, Zhejiangsheng, China) for two operation phases: standby (waiting for a measurement to be triggered) and measurement (measurement in progress). Aside from the USB serial bridge, the other component requiring the most power is the EA TFT015-22AI LCD. Its power is measured with a tabletop Keithley 2700 multimeter

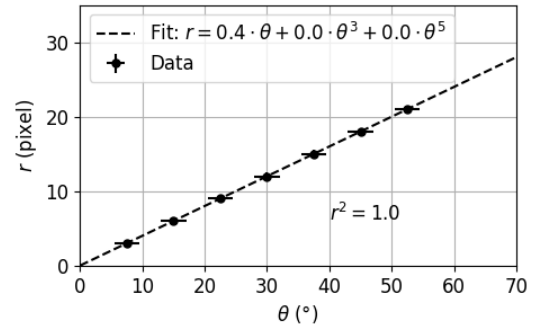


Fig. 8. Projection function of the 140° M12 lens.

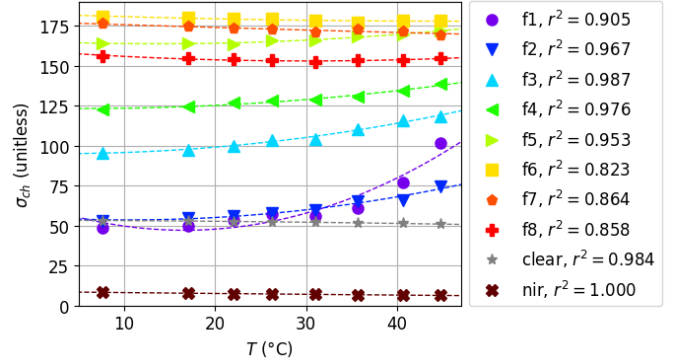


Fig. 9. LCD contrast calibration results for all channels. The linear regression result is shown as a dashed curve.

(Ohio, USA) for three operation phases: sleep (driver chip on standby and display off), display off (driver chip active and display off), and display on (driver chip active and display on).

V. RESULTS

A. Calibration

The whole calibration procedure was performed on one sensor system.

1) **Lens Projection Formula:** The projection function of the 140° lens (Fig. 8) shows that only the first order term (k_1) is non-null. The radius corresponding to the maximal angle possible for the chosen lens (70°) is thus 28 pixels. The maximum required LCD circle pattern will have a diameter of 56 pixels.

The minimal FOV distinguishable by the system is limited by the size of a pixel. The pixel size in θ is defined as the maximal angle divided by the corresponding radius in pixels, which corresponds to $2.5^\circ/\text{pixel}$. The pixel size in ϕ depends on the range of θ considered. At $\theta = 0^\circ$, the FOV is contained in a single pixel, so the pixel size in ϕ is $360^\circ/\text{pixel}$. At $\theta = 70^\circ$, the FOV is contained in a thin circle of 1 pixel thickness and a circumference of 176 pixels. The pixel size in ϕ is in that case $360^\circ/176 \text{ pixels} = 2.05^\circ/\text{pixel}$. The pixel size in ϕ of the system therefore increases with θ , whereas the pixel size in θ is constant.

2) **LCD Contrast:** The measured $\sigma_{ch}(T)$ for all channels is shown in Fig. 9. The fits show good agreement with the measured data for all channels, with r^2 between 0.823 and 1.000.

Channels $f5$, $f6$, and $f7$ have the highest contrast (respectively 163, 179, and 173 at $T = 22^\circ\text{C}$), whereas channel NIR has the lowest (7 at $T = 22^\circ\text{C}$).

These contrast values are approximately an order of magnitude lower than the expected contrast of 900 at 25°C for this LCD [46]. This can be explained by the important differences in the experimental setups, namely that the manufacturer's contrast value was measured using the back-light LED included in the package, whereas the light source used here is a light bulb situated approximately 10 cm from the LCD.

It is observed that the contrast at lower wavelengths (channels $F1$ – $F5$) increases with temperature, whereas the contrast at higher wavelengths (channel $F7$) diminishes with the temperature. This can be explained by the fact that the birefringence of a liquid crystal mixture diminishes with increasing temperature [47]. The wavelength of peak transmission is proportional to the birefringence [48], such that it also diminishes with increasing temperature. The decreasing of the wavelength of peak transmission leads to an increase in the contrast for lower wavelengths and a decrease for higher wavelengths.

The performance of the system in terms of the minimal FOV achievable (15) is therefore limited by the channels of lowest σ_{ch} , which, excluding the NIR channels, is the clear channel with $\sigma_{\text{clear}} \approx 50$ over the whole temperature range. The condition for δ_{ch} (15) becomes

$$\delta_{ch} \gg \frac{1}{49} = 0.02. \quad (21)$$

3) Delta Functions: The measured data for $\gamma_{ch,\theta}$ is shown in Fig. 10(a). The linear regression model is chosen arbitrarily based on the measured data

$$\gamma_{ch,\theta} \approx k_1 + k_2 \cdot \exp(-\theta) + k_3 \cdot \theta \cos^4 \theta \quad (22)$$

where k_1 , k_2 , and k_3 are linear regression coefficients, and θ is expressed in radian.

To fulfill the conditions listed in (12), the fit model is then normalized over $\theta \in [0^\circ, 70^\circ]$ [Fig. 10(c)]. The fits show good agreement with the measured data for all channels, with r^2 between 0.9717 and 0.9964.

The measured data for $\gamma_{ch,\phi}$ is shown in Fig. 10(b). The linear regression model is chosen arbitrarily based on the measured data

$$\gamma_{ch,\phi} = k_1 + k_2 \cdot \cos\left(\phi - \frac{\pi}{4}\right) + k_3 \cdot \cos\left(\phi - \frac{3\pi}{4}\right) \quad (23)$$

where k_1 , k_2 , and k_3 are linear regression coefficients, and ϕ is expressed in radian.

To fulfill the conditions listed in (12), the fit model is then normalized over $\phi \in [0^\circ, 360^\circ]$ [Fig. 10(d)]. $\gamma_{ch,\phi}$ was expected to be constant for all ϕ , but a periodicity is observed in the measurements. Another interesting observation is that the periodicity has two phase shifts: channels $f1$ – $f8$ share one phase shift, while channels clear and nir share another phase shift. This specific grouping of the channels indicates an alignment issue between the lens and the AS7341 light sensor because channels $f1$ – $f8$ are grouped in the center of the chip while the photodiodes for channels clear and nir

are grouped at the bottom of the active area. The photodiode array has dimensions of $520 \mu\text{m}$ by $780 \mu\text{m}$. An offset of 0.25 mm between the lens optical axis and the center of the photodiode array is possible due to the alignment tolerance of the prototype. Even though this misalignment is undesired, it can be calibrated for. The fits show good agreement with the measured data for all channels, with r^2 between 0.7820 and 0.9515.

4) PAR Model Regression: After the model was trained with all possible LVs, different optimal numbers were found for the different error metrics nRMSE [Fig. 11(a)] and MAPE [Fig. 11(b)]. The smallest nRMSE is reached for 13 LVs; the smallest MAPE is reached for the maximum number of 18 LVs. According to Geladi and Kowalski [40] not all components are normally used, as some of the smaller components probably describe noise. However, this statement could not be proven for the MAPE value, not even for different sets of new unseen data. Possible reasons for this behavior is the relatively small data set. Therefore, the maximum number of LVs was selected as a function of the smallest MAPE to achieve a low variance in the results. To evaluate the performance of the prediction, the test set is applied to the trained model. The result for a model with 18 LVs is shown in Fig. 12. An r^2 value of 0.9985 shows that the model was appropriately selected to represent the underlying data. Table III compares the performance of the present PAR sensing solution to other custom-built and commercial devices. Compared to the previous iteration of the PAR sensor node by Comella et al. [26], this work exhibits an nRMSE 2.4 times higher. This is due to the fact that less data with a high PAR was recorded in the first iteration than in the current version. By the definition of nRMSE (27), the relative error therefore also increases. But considering the present work's added features and superior regression methodology, it is a step up. In comparison to other custom sensors, the present sensor is neither the best nor the worst. Since the present method is based on calibrating the new sensor against an existing reference sensor, it could at most perform as well as the reference sensors. With a MAPE of 6.78%, this work approaches the performance of both commercial devices, which each exhibit an accuracy of 5%. It should be noted that the model has been trained with PAR values in a range up to $1000 \mu\text{mol m}^{-2}\text{s}^{-1}$. PAR under clear skies (direct sunlight) has average values of 541 – $1753 \mu\text{mol m}^{-2}\text{s}^{-1}$ depending on the sun zenith angle, while an obscured sky (diffuse sunlight) has an average PAR of $509 \mu\text{mol m}^{-2}\text{s}^{-1}$ [39]. Hence, a further validation of the model in the higher PAR range is necessary.

Table III also highlights that no other published work enables adjusting the FOV in both the θ - and ϕ -directions.

B. Characterization

1) System Spatial Resolution: The average resolution in the θ - and ϕ -directions are, respectively, $\text{FWHM}_\theta = 7^\circ$ and $\text{FWHM}_\phi = 15^\circ$ (see Fig. 13). The Nyquist theorem indicates that the minimum spatial resolution is twice the pixel size [45], which corresponds to 5° in the θ -direction (see Section V-A.1). The minimum spatial resolution in the ϕ -direction is harder to estimate through the Nyquist theorem, because the pixel

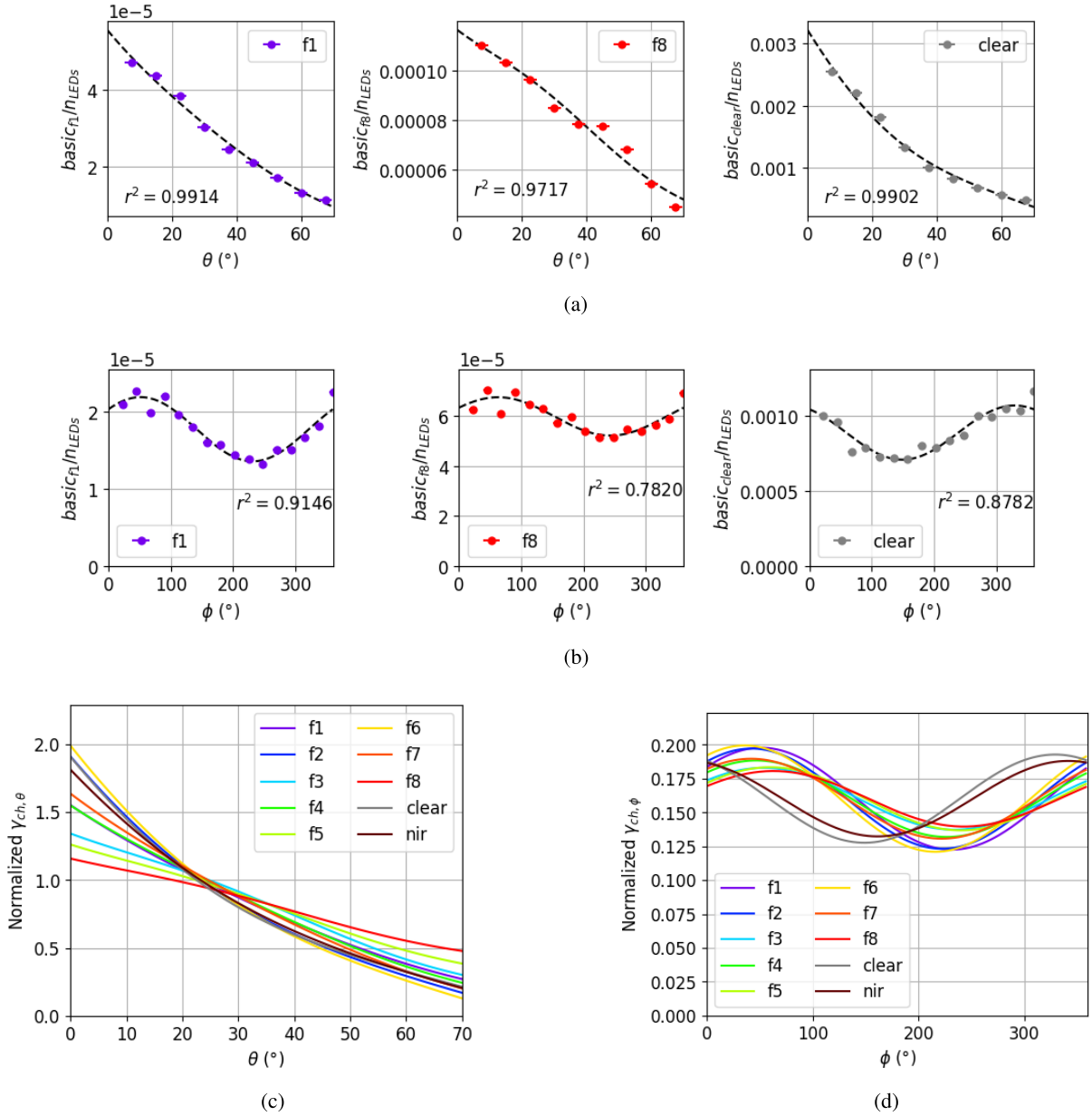


Fig. 10. Experimental determination of the $\gamma_{ch,\theta}$ and $\gamma_{ch,\phi}$ functions. Note that, to facilitate interpretation of the plots, θ and ϕ are shown here in degree, even though they are expressed in rad in the functions. (a) Experimental measurement of $\gamma_{ch,\theta}$ for channels f1, f8, and clear. The dashed curve represents the linear regression according to the model shown in (22). (b) Experimental measurement of $\gamma_{ch,\phi}$ for channels f1, f8, and clear. The dashed curve represents the linear regression according to the model shown in (23). (c) Normalized $\gamma_{ch,\theta}$. (d) Normalized $\gamma_{ch,\phi}$.

size varies depending on the θ considered. For $\theta = 35^\circ$ (half of the full possible range), the pixel size in the ϕ -direction is 4.1° , which corresponds to a minimum spatial resolution of 8.2° . The measured spatial resolution is higher (worse) in both directions. This is due to the relatively low LCD contrast ($\sigma_{clear} \approx 50$, see Fig. 9), which implies that OFF pixels do not fully block incoming light. A portion of the light incident on a given pixel is thus transmitted even if the pixel is not part of the desired FOV.

Furthermore, Fig. 13 shows that $FWHM_\theta$ and $FWHM_\phi$ exhibit a relatively high standard deviation of $\pm 2^\circ$ and $\pm 5^\circ$, respectively. This is caused by the combination of approximate LED placement within the LED sphere and pixelated nature of the FOV selected through the LCD.

While this spatial resolution would be poor for an imaging application, it is appropriate for the present application, where the goal is only to separate the sensor FOV in a small number of areas. The sensor FOV could be split in a maximum of ten sections of 7° width in the θ -direction and 24 sections of 15° width in the ϕ -direction, for a total of 240 FOV sections.

2) Power Consumption: The power consumption for the sensor prototype is measured to be 392.5 mW in the measurement state and 361.35 mW in a standby state. The measurement state power is 34 % higher than the estimated value of 292.14 mW (see Table II). The higher power consumption is attributed to a lower efficiency for the voltage converters and more than typical consumption for the FT2232H USB-to-serial bridge. While the sensor prototype's power consumption is too high

TABLE III
COMPARISON OF THE NOVEL SENSOR SYSTEM PRESENTED IN THIS WORK WITH OTHER PUBLISHED WORK AS WELL AS COMMERCIAL SENSORS. THE PERFORMANCE METRICS ARE DEFINED IN APPENDIX-B. THIS WORK'S PERFORMANCE METRICS ARE CALCULATED USING THE LI-190R AS GROUND TRUTH.

Reference	Light Sensor	Max. PAR ($\frac{\mu\text{mol}}{\text{m}^2 \text{ s}}$)	FOV $_{\theta}$	FOV $_{\phi}$	r^2 (unitless)	MAE ($\frac{\mu\text{mol}}{\text{m}^2 \text{ s}}$)	MAPE (%)	RMSE ($\frac{\mu\text{mol}}{\text{m}^2 \text{ s}}$)	nRMSE (%)
This work	AS7341	1000	[0°, 70°], adjustable	[0°, 360°], adjustable	0.999	4.14	6.78	5.51	4.82
Comella <i>et al.</i> [26], [49]	AS7341	600	[0°, 90°] ¹ , fixed	[0°, 360°], fixed	0.99			16	2
Sanchez-Azofeifa <i>et al.</i> [11]	Apogee SQ110	2250	[0°, 75°] ² , fixed	[0°, 180°], fixed				137 - 331	
Leon-Salas <i>et al.</i> [50]	AS7265x	1400	[0°, 20.5°], fixed	[0°, 360°], fixed			6.83		
Caya <i>et al.</i> [51]	VTB8440BH	1800	[0°, 50°], fixed	[0°, 360°], fixed	0.994	40.6	4.74		
Barnard <i>et al.</i> [52]	LI-190R	1700	[0°, 82°] ³ , fixed	[0°, 360°], fixed		22.0	3.49		
AccuPAR LP-80 [9]	Longitudinal probe	2500	8 segments in longitudinal direction						
AccuPAR LP-80 [9]	Apogee SQ100X	4000	[0°, 75°] ² , fixed	[0°, 360°], fixed		5 ⁴			
LI-190R [5]	LI-190R	10000	[0°, 82°] ³ , fixed	[0°, 360°], fixed		5 ⁵			

¹No information on cosine correction range.

²Range with $\pm 5\%$ directional cosine response.

³Range with valid cosine correction.

⁴Given in the datasheet as *accuracy*.

⁵Given in the datasheet as *absolute calibration, traceable to the U.S. National Institute of Standards and Technology (NIST)*.

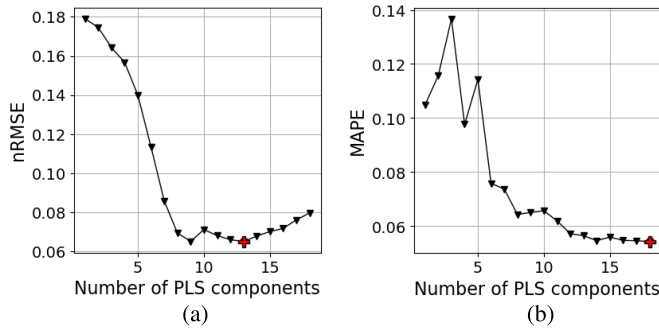


Fig. 11. Training error decreases with the number of PLS components. The smallest error is marked with a red cross. With (a) nRMSE this is reached at 13 components, with (b) MAPE at the maximum number of 18 components.

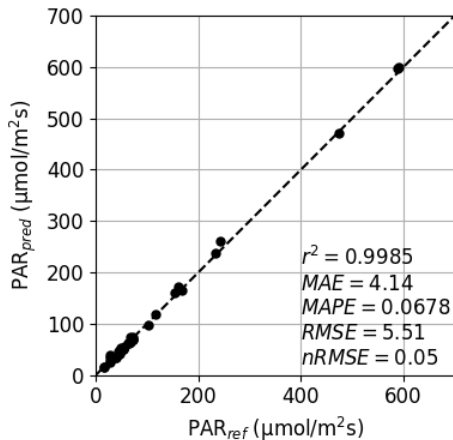


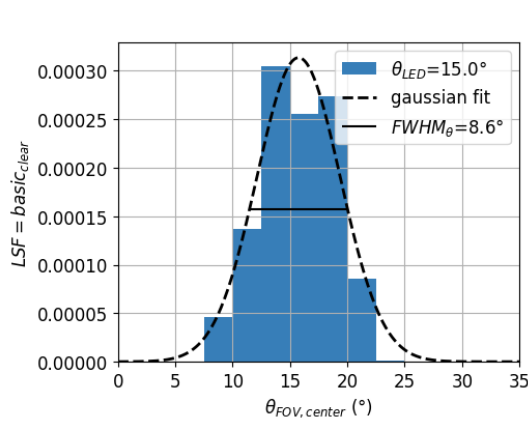
Fig. 12. Evaluation of the PAR model predictions. The calibration data ($n = 161$) was split into 75% training set ($n_{\text{training}} = 121$) and 25% test set ($n_{\text{test}} = 40$). The performance of the test set is shown here.

to be qualified as a low-power sensor, most of the power is used by the FT2232H USB-to-serial Bridge, a component that would not be used in an actual low-power implementation of the concept.

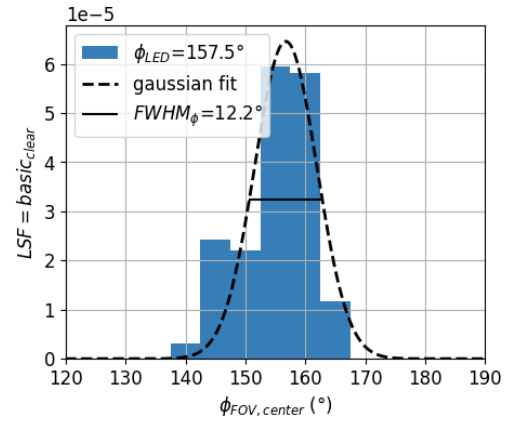
A better way to assess the low-power potential of the sensor concept is to analyze the LCD's power consumption. In the absence of the FT2232H, 37 % of the estimated power in the measurement phase would be drawn by the LCD (see Table II). The LCD power consumption is the combination of two components: the power required to actuate the liquid-crystal cells, and the power required by the LCD driver. In the sleep state, both the actuation and the driver are deactivated, which yields a low power of 0.10 mW. In the display off state, the driver is activated but no actuation is done, leading to a power consumption of 16.29 mW. Finally, in the display ON state, both the driver and the actuation are activated, leading to the maximal power consumption of 19.54 mW.

The difference in power of 3.25 mW between the “display on” and “display off” directly corresponds to the power required to drive the LCD. The pixels cover a total area of $2.772 \cdot 2.772 \text{ cm} = 7.68 \text{ cm}^2$, which leads to a power per area of $423 \mu\text{W cm}^{-2}$. This is in accordance with an approximate value of $500 \mu\text{W cm}^{-2}$ for switchable glass technology using liquid crystals [18]. This shows that most of the power required by the LCD is not used for the actual actuation, but rather for the driving circuit as well as the memory handling. Those two aspects could be improved upon using a simpler LCD technology, such as a segment display.

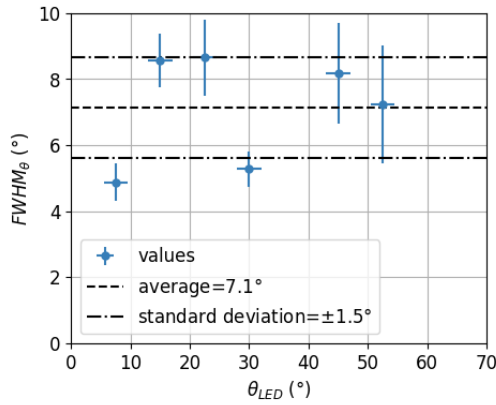
Segment displays are used in digital watches, calculators, and any device with a display meant to show a limited set of shapes and numbers/letters. In the most simple form, one of the electrodes is patterned with the desired shapes, and the other electrode (the reference) is left unpatterned and is kept at ground potential. The actuation signal consists of an ac voltage with a frequency high enough to avoid flickering and peak amplitude of up to 5 V [53]. The main drawbacks of this technology are the slow response time and the fixed patterns [33], both of which would not pose a problem for the present application. On the contrary, this actuation scheme enables much higher contrast [33], which is desirable for this application.



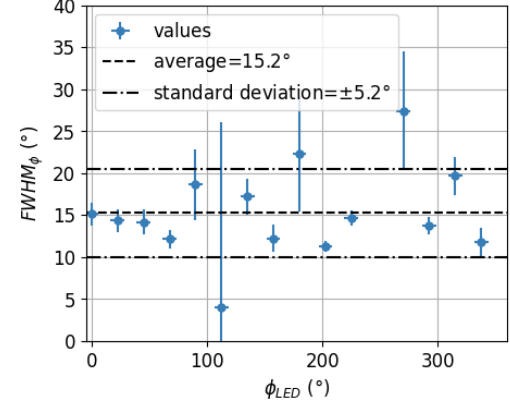
(a) Example LSF in θ for $\theta_{LED} = 15^\circ$. The measured $basic_{clear}$ are fitted with a gaussian curve from which the FWHM is calculated.



(b) Example LSF in ϕ for $\phi_{LED} = 157.5^\circ$. The measured $basic_{clear}$ are fitted with a gaussian curve from which the FWHM is calculated.



(c) Resolution in θ with respect to θ_{LED} . The horizontal error bars represent the LEDs position error and the vertical error bars represent the error on the gaussian fit.



(d) Resolution in ϕ with respect to ϕ_{LED} . The horizontal error bars represent the LEDs position error and the vertical error bars represent the error on the gaussian fit.

Fig. 13. Characterization of system's resolution in the θ - and ϕ -directions. (a) and (b) Example of the determination of the resolution via the LSF for a given θ_{LED} and ϕ_{LED} , respectively. (c) and (d) System's resolution over all θ_{LED} and ϕ_{LED} , respectively.

In a hypothetical long-term monitoring scenario, the sensor could be powered by two alkaline AA batteries in series for a combined voltage of 3 V and a capacity of 6000 mA h [54]. This corresponds to a maximal energy of 18 Wh. The present LCD could therefore be used for a total of 921 h, or more than two years if used 1 h per day. This solution is well adapted to long-term continuous monitoring.

VI. CONCLUSION

This work introduced a novel concept based on the use of an LCD as an optical shutter for intercepted PAR sensing with a dynamic FOV. A sensor prototype was assembled and calibrated, achieving a low MAE of $4.14 \mu\text{mol m}^{-2}\text{s}^{-1}$ for PAR values up to $1000 \mu\text{mol m}^{-2}\text{s}^{-1}$ and a temperature range between 4°C and 44°C . The system was able to distinguish between objects in its FOV with a spatial resolution of respectively 7° and 15° in the θ - and ϕ -directions, which is sufficient for the application. Excluding the USB-to-serial bridge used to facilitate debugging and use of the prototype, most of the power consumption is due to the LCD. It requires 19.54 mW, which could be powered by two alkaline AA batteries for more than two years for up to an hour per day. The

present system can therefore be used in long-term monitoring solutions.

This solution goes beyond the state-of-the-art sensor nodes for PAR measurements, by permitting the continuous unassisted monitoring of this parameter under different points of the tree crown with one single sensor. To the best of the authors' knowledge, no solution of this kind has been previously published. This concept based on a multispectral light sensor could also be used to measure the composition of the incident light spectrum, which has been shown to impact apple fruit development [55], [56].

A question that remains open is the stability and robustness of the sensor for long-term monitoring. The sensor system's spectral sensor and LCD can even withstand temperatures from -30°C or -20°C up to $+85^\circ\text{C}$ or $+70^\circ\text{C}$ [34], [46]. With suitable housing, humidity should also be no problem. The sensor is not expected to work under seasonal perturbations such as snow. The calibration interval is also an open question. Previous literature indicates a period of two years between sensor calibrations [57].

Future work will focus on validating the novel sensor in the field campaign for the measurement of intercepted

TABLE IV
VARIABLES FEATURED IN THE DERIVATION OF THE PAR MODEL

Parameter	Units	Description
(θ, ϕ)	-	Position in the scene coordinate system.
(r, ϕ')	-	Position in the LCD coordinate system.
$I_{\lambda,sky}(\theta, \phi)$	$\text{W m}^{-2} \text{ nm}^{-1}$	Spectral irradiance incident from the sky.
$I_{ch,sky}(\theta, \phi)$	W m^{-2}	Channel-specific irradiance incident from the sky.
$I_{\lambda,img}(r, \phi')$	$\text{W m}^{-2} \text{ nm}^{-1}$	Spectral irradiance exiting the lens, incident on the LCD.
$I_{ch,img}(r, \phi')$	W m^{-2}	Channel-specific irradiance exiting the lens, incident on the LCD.
$I_{\lambda,LCD}(r, \phi')$	$\text{W m}^{-2} \text{ nm}^{-1}$	Spectral irradiance exiting the LCD, incident on the diffuser.
$I_{ch,LCD}(r, \phi')$	W m^{-2}	Channel-specific irradiance exiting the LCD, incident on the diffuser.
$I_{\lambda,diff}$	$\text{W m}^{-2} \text{ nm}^{-1}$	Spectral irradiance exiting the diffuser, incident on the AS7341.
$I_{ch,diff}$	W m^{-2}	Channel-specific irradiance exiting the diffuser, incident on the AS7341.
$\alpha_{\lambda}(T)$	-	Lens spectral transmittance.
$\alpha_{ch}(T)$	-	Channel-specific lens transmittance.
$\xi(\theta, \phi)$	-	Lens vignetting.
$\beta_{\lambda,ON}(T)$	-	LCD spectral transmittance when all pixels are ON (transparent).
$\beta_{\lambda,OFF}(T)$	-	LCD spectral transmittance when all pixels are OFF (opaque).
$\beta_{ch,ON}(T)$	-	Channel-specific LCD transmittance when all pixels are ON (transparent).
$\beta_{ch,OFF}(T)$	-	Channel-specific LCD transmittance when all pixels are OFF (opaque).
$\sigma_{\lambda}(T)$	-	LCD spectral contrast.
$\sigma_{ch}(T)$	-	Channel-specific LCD contrast.
D	-	Total region covered by the LCD.
D_{ON}	-	LCD region where pixels are ON (transparent).
D_{OFF}	-	LCD region where pixels are OFF (opaque).
$\psi_{\lambda}(T)$	-	Diffuser spectral transmittance for light incident at the optical axis
$\psi_{ch}(T)$	-	Channel-specific diffuser transmittance for light incident at the optical axis
$\rho_{\lambda}(r, \phi')$	-	Diffuser spectral transmittance for light incident out of the optical axis
$\rho_{ch}(r, \phi')$	-	Channel-specific diffuser transmittance for light incident out of the optical axis
A_D	m^2	Total LCD area.
$\tau_{ch}(T)$	$\text{m}^2 \text{ W}^{-1}$	Light sensor irradiance responsivity.
raw_{ch}	-	Raw ADC count given by the AS7341 for a specific channel.
t_{int}	ms	AS7341 integration time.
$gain$	-	AS7341 gain.
$basic_{ch}$	-	Corrected AS7341 readout for a specific channel.
S_{λ}	-	Integral of $g_{\lambda}(\theta, \phi)$ over region D .
ω_{ch}	$\mu\text{mol W}^{-1} \text{ s}^{-1}$	Linear regression parameters that weigh the different channels $I_{ch,sky}$ to yield the PAR.
$\omega'_{ch}(T)$	$\mu\text{mol W}^{-1} \text{ s}^{-1}$	Temperature-dependent compound linear regression parameters.
$\delta_{ch}(\theta_1, \theta_2, \phi_1, \phi_2)$	-	Arbitrary function representing the proportion of the FOV currently active.

PAR and characterizing its ability to measure PAR above $1000 \mu\text{mol m}^{-2} \text{ s}^{-1}$.

APPENDIX

A. PAR Model Variables

Table IV lists all variables featured in the derivation of the PAR model (14).

B. PAR Regression Evaluation Metrics

The metrics used to evaluate the PAR regression are defined in the following list, with y_i being a real observation, $\hat{y}_i$ a predicted value, $\bar{y}$ the mean of the real observations, and n the number of observations

$$r^2 = 1 - \frac{\sum_i^n (y_i - \hat{y}_i)^2}{\sum_i^n (y_i - \bar{y})^2} \quad (24)$$

$$\text{MAE} = \frac{\sum_i^n |y_i - \hat{y}_i|}{n} \quad (25)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i^n (y_i - \hat{y}_i)^2} \quad (26)$$

$$\text{nRMSE} = \frac{\text{RMSE}}{\bar{y}} \quad (27)$$

$$\text{MAPE} = \frac{1}{n} \sum_i^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|. \quad (28)$$

C. Derivation of the Expression for $basic_{ch}$

This appendix demonstrates the derivation of the expression for $basic_{ch}$ in terms of $I_{\lambda,sky}(\theta, \phi)$ (9).

First, (5) is inserted into (6)

$$basic_{ch} = \frac{1}{A_D} \int_0^\infty \tau_{ch}(\lambda) \psi_{\lambda}(T) \cdot \left(\iint_D \rho_{\lambda}(r, \phi') \cdot I_{\lambda,LCD}(r, \phi') dA \right) d\lambda. \quad (29)$$

Then, (4) is inserted into (29). Since the regions D_{ON} and D_{OFF} are a partition of the region D , the integral over D can be expressed in the sum of the integrals over D_{ON} and D_{OFF} , respectively,

$$\begin{aligned} \text{basic}_{ch} &= \frac{1}{A_D} \int_0^\infty \tau_{ch}(\lambda) \psi_\lambda(T) \beta_{\lambda,OFF}(T) \\ &\quad \cdot \left(\sigma_\lambda(T) \iint_{D_{ON}} \rho_\lambda(r, \phi') I_{\lambda,img}(r, \phi') dA \right. \\ &\quad \left. + \iint_{D_{OFF}} \rho_\lambda(r, \phi') I_{\lambda,img}(r, \phi') dA \right) d\lambda. \end{aligned} \quad (30)$$

As stated previously, $\{D_{ON}, D_{OFF}\}$ is a partition of D , such that the following property applies for a sum of integrals [58]:

$$\begin{aligned} S_\lambda &= \iint_D \rho_\lambda(r, \phi') I_{\lambda,img}(r, \phi') dA \\ &= \iint_{D_{ON}} \rho_\lambda(r, \phi') I_{\lambda,img}(r, \phi') dA \\ &\quad + \iint_{D_{OFF}} \rho_\lambda(r, \phi') I_{\lambda,img}(r, \phi') dA. \end{aligned} \quad (31)$$

This enables expressing the integral over the region D_{OFF} from (30) in terms of the constant S_λ and the integral over D_{ON}

$$\begin{aligned} &\iint_{D_{OFF}} \rho_\lambda(r, \phi') I_{\lambda,img}(r, \phi') dA \\ &= S_\lambda - \iint_{D_{ON}} \rho_\lambda(r, \phi') I_{\lambda,img}(r, \phi') dA. \end{aligned} \quad (32)$$

Inserting (32) into (30) yields

$$\begin{aligned} \text{basic}_{ch} &= \frac{1}{A_D} \int_0^\infty \tau_{ch}(\lambda) \psi_\lambda(T) \beta_{\lambda,OFF}(T) S_\lambda \\ &\quad \cdot \left(1 + \frac{\sigma_\lambda(T) - 1}{S_\lambda} \iint_{D_{ON}} \rho_\lambda(r, \phi') I_{\lambda,img}(r, \phi') dA \right) d\lambda. \end{aligned} \quad (33)$$

Equation (3) can then be inserted into (33), giving an expression for basic_{ch} in terms of $I_{\lambda,sky}(\theta, \phi)$

$$\begin{aligned} \text{basic}_{ch} &= \frac{1}{A_D} \int_0^\infty \tau_{ch}(\lambda) \psi_\lambda(T) \beta_{\lambda,OFF}(T) S_\lambda \alpha_\lambda(T) \\ &\quad \cdot \left(1 + \frac{\sigma_\lambda(T) - 1}{S_\lambda} \iint_{D_{ON}} \rho_\lambda(r, \phi') \xi(\theta, \phi) \right. \\ &\quad \left. I_{\lambda,sky}(\theta, \phi) dA \right) d\lambda. \end{aligned} \quad (34)$$

At this point, both world (θ, ϕ) and LCD (r, ϕ') coordinate systems are present. It is desired to use only the world coordinate system. The coordinate transformation is performed based on the lens projection formula (1) and the image inverting property of lenses (2). The infinitesimal area dA on the LCD plane thus becomes

$$\begin{aligned} dA &= r \cdot dr \cdot d\phi' \\ &= r(\theta) \cdot r'(\theta) d\theta \cdot d\phi \end{aligned} \quad (35)$$

and the diffuser spectral transmittance $\rho_\lambda(r, \phi')$ becomes

$$\rho_\lambda(r, \phi') = \rho_\lambda(r(\theta), \phi'(\phi)) = \rho_\lambda(\theta, \phi). \quad (36)$$

Equations (35) and (36) are inserted into (34)

$$\begin{aligned} \text{basic}_{ch} &= \frac{1}{A_D} \int_0^\infty \tau_{ch}(\lambda) \psi_\lambda(T) \beta_{\lambda,OFF}(T) S_\lambda \alpha_\lambda(T) \\ &\quad \cdot \left[1 + \frac{\sigma_\lambda(T) - 1}{S_\lambda} \iint_{D_{ON}} (I_{\lambda,sky}(\theta, \phi) \right. \\ &\quad \left. \cdot \rho_\lambda(r(\theta), \phi'(\phi)) \xi(\theta, \phi) r(\theta) r'(\theta) d\theta d\phi) \right] d\lambda. \end{aligned} \quad (37)$$

Finally, the term inside the integral over D_{ON} is defined as the positive function $g_\lambda(\theta, \phi)$

$$g_\lambda(\theta, \phi) = \rho_\lambda(r(\theta), \phi'(\phi)) \xi(\theta, \phi) I_{\lambda,sky}(\theta, \phi) r(\theta) r'(\theta). \quad (38)$$

Inserting (38) into (37) yields

$$\begin{aligned} \text{basic}_{ch} &= \frac{1}{A_D} \int_0^\infty S_\lambda \tau_{ch}(\lambda) \psi_\lambda(T) \beta_{\lambda,OFF}(T) \alpha_\lambda(T) \\ &\quad \cdot \left(1 + \frac{\sigma_\lambda(T) - 1}{S_\lambda} \iint_{D_{ON}} g_\lambda(\theta, \phi) d\theta d\phi \right) d\lambda. \end{aligned} \quad (39)$$

Equation (39) is the exact expression of basic_{ch} in terms of $I_{\lambda,sky}(\theta, \phi)$.

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